

The Rising Sea - Self Study

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Abstract

This document contains notes from my reading of Ravi Vakil's *The Rising Sea* ([Vak24]). I read this book with my friend, [Ophelia You](#), over the course of the summer of 2026. Our goal was to gain a working understanding of algebraic geometry, since it was becoming increasingly applicable in both of our fields of interest.

I have personally already encountered (pre)sheaves, sheaf cohomology via the derived functors of Γ , Čech cohomology, affine varieties, (pre)cosheaves, and the notion of a structure sheaf on a space, largely due to my spring 2026 classes in complex geometry and factorization algebras (see [Tal26a] and [Tal26b], respectively). I also have experience with category theory, abelian categories, and some exposure to homological algebra, due to these same classes and my readings on [ML98, Lur09, Hov99, Rie14, Alu09, Wei94].

As such, I'm approaching the book looking to solidify the various holes in this background, as well as to gain knowledge of schemes and to deepen my understanding of algebraic geometry. For these reasons, some parts of these notes will seem very barren and empty, possibly not even recording the important definitions and theorems in the book, and only writing a couple things. This is intentional – I see no reason to write down the definition of a functor for the 100th time when it is a concept I have been confidently using for the past year and a half. Additionally, I am taking an algebraic geometry class in the fall, taught by professor [Matiej Szczesny](#), so I will be treating this reading as preparation for that class. Any notes not credited to a bibliography source are either directly from [Vak24], are my interpretation of a passage therein, or are directly from me.

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Part I

Preliminaries

Chapter 1

Just enough category theory to be dangerous

1.1 Categories and functors

Exercise 1.1.1. Make sense of the definition of a group as a groupoid with one object. Then describe a groupoid that is not a group.

Proof. For the first, given a group G , we define a category $\mathbf{BG}$ with one object $*$, morphisms $\mathbf{BG}(*, *) := G$, and composition, for any $g, h \in \mathbf{BG}(*, *)$, given by $h \circ g := h \cdot g$, for $\cdot$ the group operation. For the second, let X a space, and $\Pi_1(X)$ the fundamental groupoid given by taking $\text{Obj}(\Pi_1(X)) := X$ and, for any $x, y \in X$, define $\Pi_1(X)(x, y)$ to be homotopy classes of paths from x to y . The subscript 1 is a reference to the fact that one could define a fundamental n -groupoid for any n , including $n = \infty$. Often we simply write $\Pi(X)$ for $\Pi_1(X)$. ■

Notation 1.1.2. Let $\mathcal{C}$ a category. If $x, y \in \mathcal{C}$, then [Vak24] often writes $h^c : \mathcal{C} \rightarrow \text{Set}$ for the representable functor $h^c = \mathcal{C}(c, -)$. Similarly, he writes $h_c = \mathcal{C}(-, c) : \mathcal{C}^{\text{op}} \rightarrow \text{Set}$. He refers to the latter as the *functor of points*.

Exercise 1.1.3. Let $\mathbf{fVect}_k$ be the category of finite-dimensional k -vector spaces. Show that the double dual functor $(-)^{**} \cong 1 : \mathbf{fVect}_k \rightarrow \mathbf{fVect}_k$.

Proof. Let V a finite-dimensional k -vector space. Let $v \in V$, let $\varphi \in \text{Hom}_k(V, k)$, and write $\varepsilon_v : \text{Hom}_k(V, k) \rightarrow k$ for the map $\varepsilon_v(\varphi) = \varphi(v)$. We claim the map $v \mapsto \varepsilon_v$ determines a natural isomorphism $\varepsilon : (-)^{**} \cong 1$. First, note that this map is k -linear: for $u, v \in V$, by k -linearity of φ ,

$$\varepsilon_{u+v}(\varphi) = \varphi(u + v) = \varphi(u) + \varphi(v) = \varepsilon_u(\varphi) + \varepsilon_v(\varphi).$$

We show this is an isomorphism by counting dimensions. We are allowed to choose a basis to prove that ε is an isomorphism without making it non-canonical, since the map itself is not canonical. Let $\{e_i\}_{1 \leq i \leq n} \subseteq V$ a basis, let $\{\delta_i\}_{1 \leq i \leq n} \subseteq V^*$ be the dual basis $\delta_i(e_j) = \delta_{ij}$, and let $\{\varepsilon_i\}_{1 \leq i \leq n} \subseteq V^{**}$ be $\varepsilon_i := \varepsilon_{e_i}$. Of course it is easy to show that the dual basis δ_i is indeed a basis. Given $\varphi \in V^*$, we immediately have

$$\varphi = \sum_{1 \leq i \leq n} \varphi(e_i) \delta_i.$$

Moreover, the δ_i 's are linearly independent by linear independence of the e_i 's: if $\{c_i\}_{1 \leq i \leq n} \subseteq k$ such that $\sum_i c_i \delta_i = 0$, then

$$\left(\sum_{1 \leq i \leq n} c_i \delta_i \right) (e_j) = c_j \delta_j(e_j) = c_j = 0$$

implies $c_j = 0$ for all j . Now simply note that $\varepsilon_i(\delta_j) = \delta_j(e_i) = \delta_{ij}$, meaning $\{\varepsilon_i\}_{1 \leq i \leq n}$ is the dual basis of $\{\delta_i\}_{1 \leq i \leq n}$. Thus $\dim V = \dim V^* = \dim V^{**}$. Moreover, we see that the map $\varepsilon_{(-)}$ maps the basis $\{e_i\}_{1 \leq i \leq n}$ to the basis $\{\varepsilon_i\}_{1 \leq i \leq n}$, meaning it is an isomorphism.

Finally, we show naturality. Let $\varphi : V \rightarrow W$ be k -linear, and $v \in V$. Since $(-)^* = \text{Hom}_k(-, k)$, the induced map $\varphi^* : V^* \rightarrow W^*$ is given by precomposition by φ , and the induced map $\varphi^{**} = (\varphi^*)^* : V^{**} \rightarrow W^{**}$ is given by precomposition by φ^* . Naturality amounts to showing $\varphi^{**}(\varepsilon_v) = \varepsilon_{\varphi(v)}$. Let $\psi \in V^*$. We have

$$\varphi^{**}(\varepsilon_v)(\psi) = (\varepsilon_v \circ \varphi^*)(\psi) = \varepsilon_v(\psi \circ \varphi) = (\psi \circ \varphi)(v) = \psi(\varphi(v)) = \varepsilon_{\varphi(v)}(\psi).$$

This proves naturality. ■

1.2 Universal properties determine an object up to unique isomorphism

Example 1.2.1. Let A a ring and $S \subseteq A$ a subset closed under multiplication and containing 1. Such a subset is called *multiplicative*. The *localization of A at S* , denoted $S^{-1}A$, is initial with respect to inverting S . More precisely, given any A -algebra B and algebra homomorphism $A \rightarrow B$ which sends each $s \in S$ to an invertible element of B must factor uniquely through a map $A \rightarrow S^{-1}A$. We generally define $S^{-1}A$ by formally adding inverses $1/s$ to each $s \in S$, and we define the map by $a \mapsto a/1$.

Remark 1.2.2. One can also define localizations of A -modules at a multiplicative subset S of A .

Exercise 1.2.3. Let A, B, C rings and $A \rightarrow B, A \rightarrow C$ maps of rings. Define a ring structure on $B \otimes_A C$ in a natural way.

Proof. First we define an A -module structure on B and C . Let $\varphi_B : A \rightarrow B$ and $\varphi_C : A \rightarrow C$ be the given maps, and define for any $a \in A, b \in B, c \in C$ scalar multiplication by $a \cdot b := \varphi_B(a)b$, and $a \cdot c := \varphi_C(a)c$. Since φ_B and φ_C are ring homomorphisms, this is easily seen to yield A -module structures on the given rings. We thus have an A -module $B \otimes_A C$, so we define multiplication in the natural way on pure tensors by

$$(b_1 \otimes c_1)(b_2 \otimes c_2) := b_1 b_2 \otimes c_1 c_2.$$

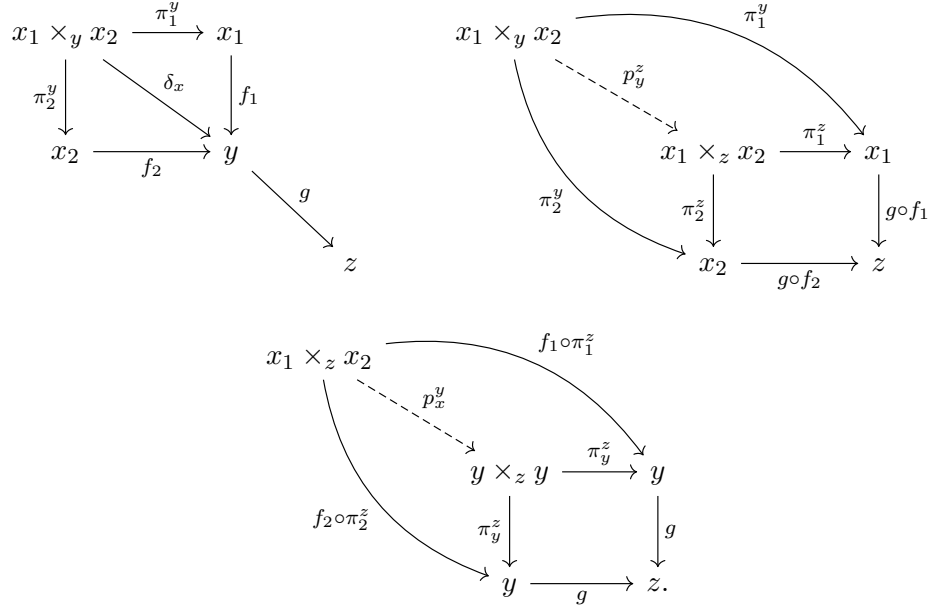
Definition 1.2.4. If $\pi : x \rightarrow y$ is a morphism and $x \times_y x$ exists, then the map $\pi \times_y \pi : x \rightarrow x \times_y x$ is denoted δ_π , and called the *diagonal morphism*. ■

Exercise 1.2.5. Let $x_1, x_2 \rightarrow y$ and $y \rightarrow z$ be morphisms. Show that there is a pullback diagram

$$\begin{array}{ccc} x_1 \times_y x_2 & \longrightarrow & x_1 \times_z x_2 \\ \downarrow & & \downarrow \\ y & \longrightarrow & y \times_z y \end{array}$$

Proof. To make the proof easier to read, the following diagrams establish notation for all relevant

maps, and make it transparent how the map $p_y^z : x_1 \times_y x_2 \rightarrow x_1 \times_z x_2$ arises:



We label these diagrams (1), (2), and (3), respectively. Our goal is to show that the diagram

$$\begin{array}{ccc}
 x_1 \times_y x_2 & \xrightarrow{p_y^z} & x_1 \times_z x_2 \\
 \delta_x \downarrow & & \downarrow p_x^y \\
 y & \xrightarrow{\delta_g} & y \times_z y
 \end{array}$$

commutes, and is a pullback diagram. By universality, $p_x^y \circ p_y^z : x_1 \times_y x_2 \rightarrow y \times_z y$ should give rise to a morphism $p_x^y \circ p_y^z : x_1 \times_y x_2 \rightarrow y$ as indicated by the dotted arrows, rendering the diagram

$$\begin{array}{ccc}
 x_1 \times_y x_2 & \xrightarrow{p_x^y \circ p_y^z} & y \times_z y \\
 & \searrow & \downarrow \pi_y^z \\
 & & y \xrightarrow{g} z
 \end{array}$$

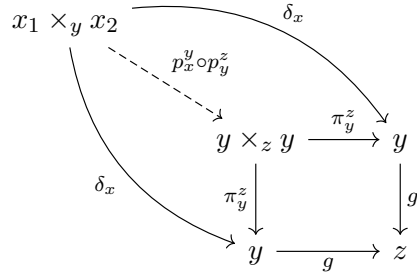
commute. Commutativity forces the dotted arrows to be $\pi_y^z \circ p_x^y \circ p_y^z$. Using diagrams (1-3), we find that for any $i = 1, 2$,

$$\pi_y^z \circ p_x^y \circ p_y^z = f_i \circ \pi_i^z \circ p_y^z \quad (3)$$

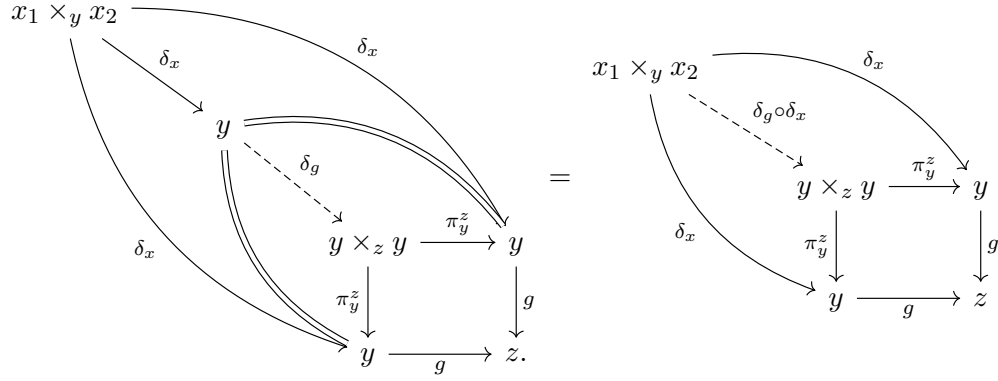
$$= f_i \circ \pi_i^y \quad (2)$$

$$= \delta_x \quad (1).$$

Hence

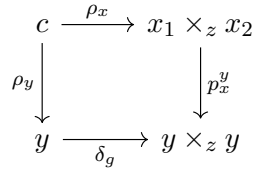


commutes. Now consider the diagram

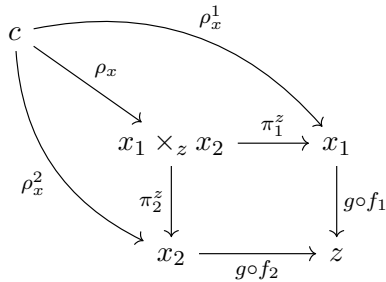


By uniqueness of the dashed arrows, we conclude that $\delta_g \circ \delta_x = p_x^y \circ p_y^z$.

Now we show that the given square is a pullback. Let $c \in \mathcal{C}$, let $\rho_x : c \rightarrow x_1 \times_z x_2$, and let $\rho_y : c \rightarrow y$ such that the diagram



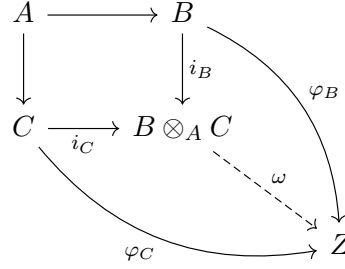
commutes. Looking at ρ_x in particular, universality of pullback means that there exist morphisms ρ_x^i , $i = 1, 2$, rendering the diagram



commutative. ■

Exercise 1.2.6. Let $A \rightarrow B$ and $A \rightarrow C$ be ring maps. Show that there are natural maps $B \rightarrow B \otimes_A C$ and $C \rightarrow B \otimes_A C$ such that $B \otimes_A C$ is the pushout $B \amalg_A C$ in $\mathcal{Alg}_A$ along these maps.

Proof. Let $i_B : B \rightarrow B \otimes_A C$ be the map $b \mapsto b \otimes 1$, and let $i_C : C \rightarrow B \otimes_A C$ be defined the same. Let $\varphi_B : B \rightarrow Z$ and $\varphi_C : C \rightarrow Z$ be maps rendering the diagram of solid arrows



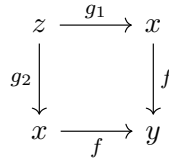
commutative. We want to show there is a unique way to define ω such that the diagram commutes. By commutativity of the top triangle, we must have $\omega(b \otimes 1) = \varphi_B(b)$ for all $b \in B$, and likewise by commutativity of the bottom triangle we have $\omega(1 \otimes c) = \varphi_C(c)$ for $c \in C$. We then recall the ring structure in exercise 1.2.3 to write $b \otimes c = (b \otimes 1)(1 \otimes c)$. Since ω must be a ring homomorphism, we define

$$\omega(b \otimes c) = \omega(b \otimes 1)\omega(1 \otimes c) = \varphi_B(b)\varphi_C(c).$$

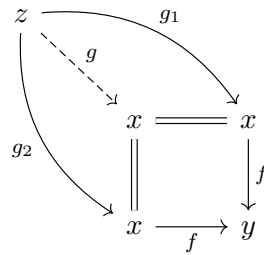
This makes sense since multiplication is a bilinear map, and tensor products are universal with respect to being bilinear. Finally, ω is well-defined as an A -algebra map since φ_B and φ_C agree on the A -action. ■

Exercise 1.2.7. Show that $f : x \rightarrow y$ is a monomorphism if and only if $x \times_y x$ exists and $\delta_f : x \rightarrow x \times_y x$ is an isomorphism.

Proof. ($\implies$) Let f be mono. Let $g_1, g_2 : z \rightarrow x$ be maps such that

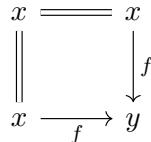


commutes. Since f is mono, $g_1 = g_2$, so there exists a unique morphism $g = g_1 = g_2$ making the diagram



commutes. Hence the given cone $x = x = x$ is a pullback. The cone that gives rise to δ_f is the cone $x = x = x$, so it follows that $\delta_f = 1_x$, hence δ_f is an isomorphism.

($\impliedby$) Let $x \times_y x$ exist and $\delta_f : x \cong x \times_y x$. Since δ_f is an isomorphism, this induces an isomorphism of cones, meaning



is a pullback square. Let $g_1, g_2 : z \rightarrow x$ such that $f \circ g_1 = f \circ g_2$. We can display this information as a diagram

$$\begin{array}{ccc} z & \xrightarrow{g_1} & x \\ g_2 \downarrow & & \downarrow f \\ x & \xrightarrow{f} & y. \end{array}$$

Universality of pullback induces a unique dashed arrow g as indicated:

$$\begin{array}{ccccc} z & & & \xrightarrow{g_1} & x \\ & \searrow \text{dashed } g & & & \downarrow f \\ & x & \xrightarrow{=} & x & \\ & \parallel & & & \downarrow f \\ & x & \xrightarrow{f} & y & \end{array}$$

rendering the diagram commutative. We thus see $g = g_1 = g_2$. Therefore, $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$, meaning f is mono. ■

Exercise 1.2.8. Let $f : y \rightarrow z$ be mono, and $x_1, x_2 \in \mathcal{C}$. Show that the map $x_1 \times_y x_2 \rightarrow x_1 \times_z x_2$ is an isomorphism. (Hint: prove $\mathcal{C}(-, x_1 \times_y x_2) \cong \mathcal{C}(-, x_1 \times_z x_2)$. The statement follows since Yoneda reflects isomorphisms).

Proof. Let

$$x_1 \xrightarrow{g_1} y \xleftarrow{g_2} x_2$$

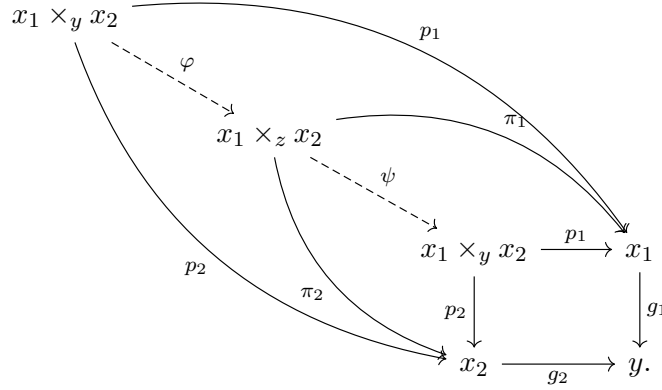
be the span diagram. Note that the map $\varphi : x_1 \times_y x_2 \rightarrow x_1 \times_z x_2$ arises as

$$\begin{array}{ccccc} x_1 \times_y x_2 & & & \xrightarrow{p_1} & x_1 \\ & \searrow \text{dashed } \varphi & & & \downarrow f \circ g_1 \\ & x_1 \times_z x_2 & \xrightarrow{\pi_1} & x_1 & \\ & \downarrow \pi_2 & & & \downarrow f \circ g_2 \\ & x_2 & \xrightarrow{f \circ g_2} & z. & \end{array}$$

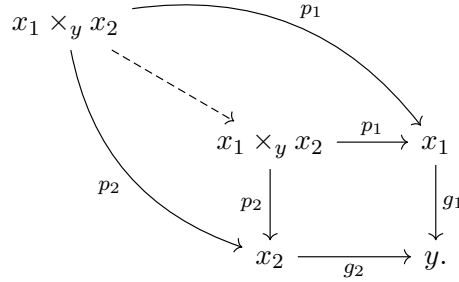
This diagram expresses the equalities $f \circ g_2 \circ \pi_2 = f \circ g_1 \circ \pi_1$ and $f \circ g_2 \circ p_2 = f \circ g_1 \circ p_1$. Since f is mono, $g_2 \circ \pi_2 = g_1 \circ \pi_1$ and $g_2 \circ p_2 = g_1 \circ p_1$. In particular, the diagram

$$\begin{array}{ccccc} x_1 \times_y x_2 & & & \xrightarrow{p_1} & x_1 \\ & \searrow \text{dashed } \varphi & & & \downarrow g_1 \\ & x_1 \times_z x_2 & \xrightarrow{\pi_1} & x_1 & \\ & \downarrow \pi_2 & & & \downarrow g_1 \\ & x_2 & \xrightarrow{g_2} & y & \\ & & & \downarrow f & \\ & & & z & \end{array}$$

commutes. In particular, we get a map $\psi : x_1 \times_z x_2 \rightarrow x_1 \times_y x_2$ via universality:



Note that the inner 2 squares commute by universality of $x_1 \times_y x_2$, and the outer 2 squares commute since they are the above diagram. Hence $\psi \circ \varphi$ satisfies the dashed arrow in the diagram



However, the identity arrow also suffices as this dashed arrow, and by universality we have $\psi \circ \varphi = 1$. We may similarly arrange the squares in the reverse order to see that $\varphi \circ \psi = 1$. Thus φ is an isomorphism. $\blacksquare$

Exercise 1.2.9. 1. Let $a, a' \in \mathcal{C}$, and maps $i_c : \mathcal{C}(c, a) \rightarrow \mathcal{C}(c, a')$ for all $c \in \mathcal{C}$, commuting with all precomposition maps in the sense that if $f : b \rightarrow c$, then

$$\begin{array}{ccc} \mathcal{C}(c, a) & \xrightarrow{i_c} & \mathcal{C}(c, a') \\ f^* \downarrow & & \downarrow f^* \\ \mathcal{C}(b, a) & \xrightarrow{i_b} & \mathcal{C}(b, a') \end{array}$$

commutes. Show that there is a unique morphism $g : a \rightarrow a'$ which induces the i_c 's.

2. If each i_c is a bijection, show that g is an isomorphism.

Proof. (1) Let $c \in \mathcal{C}$ and $f \in \mathcal{C}(c, a)$. We will show that $i_c(f) = i_a(1_a) \circ f$, and will define $g := i_a(1_a)$. This is sufficient to prove the statement since c is an arbitrary object and f is an arbitrary element of the domain of i_c . Since $f \in \mathcal{C}(c, a)$, we have $f : c \rightarrow a$, and thus $f^* : \mathcal{C}(a, a) \rightarrow \mathcal{C}(c, a)$. Thus we consider the diagram

$$\begin{array}{ccc} \mathcal{C}(a, a) & \xrightarrow{i_a} & \mathcal{C}(a, a') \\ f^* \downarrow & & \downarrow f^* \\ \mathcal{C}(c, a) & \xrightarrow{i_c} & \mathcal{C}(c, a'), \end{array}$$

and note that 1_a is an element of $\mathcal{C}(a, a)$. We chase it around the diagram:

$$\begin{array}{ccc} 1_a & \xrightarrow{i_a} & i_a(1_a) \\ f^* \downarrow & & \downarrow f^* \\ f & \xrightarrow{i_c} & i_c(f) = i_a(1_a) \circ f. \end{array}$$

The two equalities in the bottom right are given by chasing 1_a around either direction of the diagram. Since the diagram commutes by hypothesis, the equality is forced.

(2) If each i_c is a bijection, then in particular $i_{a'}$ is a bijection, meaning there exists $i_{a'}^{-1} : \mathcal{C}(a', a') \rightarrow \mathcal{C}(a', a)$. We can consider the map $i_{a'}(1_{a'}) : a' \rightarrow a$. We want to show that this map is an inverse to $i_a(1_a)$. To do this, we start by considering the diagram

$$\begin{array}{ccc} \mathcal{C}(a, a) & \xrightarrow{i_a} & \mathcal{C}(a, a') \\ i_{a'}^{-1}(1_{a'})^* \downarrow & & \downarrow i_{a'}^{-1}(1_{a'})^* \\ \mathcal{C}(a', a) & \xrightarrow{i_{a'}} & \mathcal{C}(a', a'). \end{array}$$

We can chase 1_a around this diagram:

$$\begin{array}{ccc} 1_a & \xrightarrow{i_a} & i_a(1_a) \\ i_{a'}^{-1}(1_{a'})^* \downarrow & & \downarrow i_{a'}^{-1}(1_{a'})^* \\ i_{a'}^{-1}(1_{a'}) & \xrightarrow{i_{a'}} & i_{a'}(i_{a'}^{-1}(1_{a'})) = i_a(1_a) \circ i_{a'}^{-1}(1_{a'}). \end{array}$$

Note that $i_{a'}(i_{a'}^{-1}(1_{a'})) = 1_{a'}$ since $i_{a'} \circ i_{a'}^{-1} = 1$, thus we have $i_a(1_a) \circ i_{a'}^{-1}(1_{a'}) = 1_{a'}$.

For the other direction, note that rules of commutative diagrams allow us to swap the i_c 's for i_c^{-1} 's in the commutative diagrams. For a more rigorous proof of this, let $f : b \rightarrow c$. Then

$$\begin{aligned} f^* \circ i_c &= i_b \circ f^* \\ \implies f^* &= i_b \circ f^* \circ i_c^{-1} \\ \implies i_b^{-1} \circ f^* &= f^* \circ i_c^{-1}. \end{aligned}$$

Thus we can consider the diagram

$$\begin{array}{ccc} \mathcal{C}(a', a') & \xrightarrow{i_{a'}^{-1}} & \mathcal{C}(a', a) \\ i_a(1_a)^* \downarrow & & \downarrow i_a(1_a)^* \\ \mathcal{C}(a, a') & \xrightarrow{i_a^{-1}} & \mathcal{C}(a, a) \end{array}$$

and chase $1_{a'}$ around the diagram:

$$\begin{array}{ccc} 1_{a'} & \xrightarrow{i_{a'}^{-1}} & i_{a'}^{-1}(1_{a'}) \\ i_a(1_a)^* \downarrow & & \downarrow i_a(1_a)^* \\ i_a(1_a) & \xrightarrow{i_a^{-1}} & i_a^{-1}(i_a(1_a)) = i_{a'}^{-1}(1_{a'}) \circ i_a(1_a) \end{array}$$

Since $i_a^{-1}(i_a(1_a)) = 1_a$, we have that $i_{a'}^{-1}(1_{a'}) \circ i_a(1_a) = 1_a$. Therefore $i_a(1_a)^{-1} = i_{a'}^{-1}(1_{a'})$. ■

Exercise 1.2.10. 1. Let $a, b \in \mathcal{C}$. Prove that $\mathcal{C}(b, a) \cong \text{Nat}(\mathcal{C}(a, -), \mathcal{C}(b, -))$.

2. State and prove the corresponding fact for contravariant functors.

3. Let $F : \mathcal{C} \rightarrow \text{Set}$. Prove $F(a) \cong \text{Nat}(\mathcal{C}(a, -), F)$, natural in a .

Proof. (1) It suffices to show that the only maps $\eta : \mathcal{C}(a, -) \Rightarrow \mathcal{C}(b, -)$ are given by precomposition. Note that in ?? 1.2.9, we showed that if $i : \mathcal{C}(-, a) \Rightarrow \mathcal{C}(-, b)$, then i was given by postcomposition by a particular morphism. The statement thus follows by duality.

(2) The statement is that $\text{Nat}(\mathcal{C}(-, a), \mathcal{C}(-, b)) \cong \mathcal{C}(a, b)$. This was proven in ?? 1.2.9.

(3) The proof is the same as in ?? 1.2.9: you let $\eta : \mathcal{C}(a, -) \Rightarrow F$, let $f \in \mathcal{C}(a, b)$, and trace $1_a \in \mathcal{C}(a, a)$ around the relevant naturality diagram to find that $\eta_b(f) = \eta_a(1_a) \circ F(f)$. This yields the given isomorphism. Naturality requires a bit more pushing, but it follows from naturality of pre- and post-composition. ■

1.3 Limits and colimits

Exercise 1.3.1. Let $\mathcal{J}$ be a poset with initial object e . Show that for any $F : \mathcal{J} \rightarrow \mathcal{C}$, $\lim F$ exists.

Proof. I prefer $0 \in \mathcal{J}$ for the initial object. Let $F : \mathcal{J} \rightarrow \mathcal{C}$. We show that $\lim F \cong F(0)$. The fact that $F(0)$ is a cone is immediate, since for any $i \in \mathcal{J}$, we have $0 \leq i$ by definition, and thus there is $F(0 \leq i) : F(0) \rightarrow F(i)$. Now let $\eta : \Delta(c) \Rightarrow F$ be another cone, where $c \in \mathcal{C}$ and $\Delta : \mathcal{C} \rightarrow \mathcal{C}^{\mathcal{J}}$ is the diagonal functor. Then in particular there is a map $\eta_0 : c \rightarrow F(0)$, and manifestly this map is the unique map making the relevant diagram commute. ■

Example 1.3.2. Let A be a ring. The ring of formal power series, denoted $A[[x]]$, is described as the A -algebra generated by a generating set $\{x^n \mid n \geq 0\}$, subject to the expected relations. One can describe it as the limit $\lim F$, where $F : \omega^{\text{op}} \rightarrow \text{Ring}$ maps $n \mapsto A[x]/(x^{n+1})$.

exercise 1.3.D because ophelia couldnt do it: Interpret the union of some subsets as a colimit.

Proof. Let X be a set and $\{U_i \subseteq X\}_{i \in I}$ a collection of subsets. Let $\mathcal{U}$ be the full subcategory of the power set $\mathcal{P}(X)$ spanned by $\{U_i\}_{i \in I}$. Let $F : \mathcal{U} \rightarrow \text{Set}$ act on objects by $U_i \mapsto U_i$, and act on morphisms by mapping subsets $U_i \subseteq U_j$ to the inclusion map $U_i \hookrightarrow U_j$. Then $\bigcup_i U_i \cong \text{colim } F$. One may alternatively define this as a colimit in the poset category $\mathcal{P}(X)$ by noting that there is an inclusion functor $i : \mathcal{U} \subseteq \mathcal{P}(X)$, and $\bigcup_i U_i \cong \text{colim } i$. One may also define G to be the evident inclusion functor $\mathcal{P}(X) \subseteq \text{Set}$, and $\bigcup_i U_i \cong \text{colim}(G \circ i)$. We usually write this as $\text{colim}(G \circ i) = \text{colim}_{i \in I} G(U_i)$; this notation specifies the subcategory $\mathcal{U}$ indexed by I using indices rather than the inclusion morphism i , and is often very convenient. ■

$$\tau_A : \text{Hom}_{\mathcal{B}}(F(A), -) \rightarrow \text{Hom}_{\mathcal{A}}(A, G(-))$$

1.4 Adjoints

1.5 Abelian categories

Exercise 1.5.1 (FHHF). Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a functor of abelian categories, and $x^\bullet \in \text{Ch}_{\mathcal{A}}$.

1. If F is right-exact, then there is a map $H^\bullet \circ F \Rightarrow F \circ H^\bullet$.

2. If F is left-exact, then there is a map $H^\bullet \circ F \Rightarrow F \circ H^\bullet$.
3. If F is exact, then $F \circ H^\bullet \cong H^\bullet \circ F$.

Proof.

■

Chapter 2

Sheaves

2.1 Motivating example: the sheaf of smooth functions

Let X be a smooth manifold. Given $U \subseteq X$ open, denote the set $C^\infty(U)$ by $\mathcal{O}(U)$. Given $U \subseteq V$, precomposition by $U \hookrightarrow V$ induces a restriction morphism. Note that $\mathcal{O}(U)$ is a ring, using the ring structure of $\mathbb{R}$ to define addition and multiplication pointwise. Thus $\mathcal{O} : \text{Open}(X)^{\text{op}} \rightarrow \text{Ring}$. Such a structure is called a *presheaf*.

Let $\{U_i \subseteq U\}_{i \in I}$ be an open cover, and let $\{f_i \in \mathcal{O}(U_i)\}_{i \in I}$ a collection of functions such that, for all $i, j \in I$, we have $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Then one can glue together the f_i 's into a map $f \in \mathcal{O}(U)$ via $f|_{U_i} = f_i$, and moreover f is the unique map with this property. This property makes $\mathcal{O}$ into a *sheaf*.

2.2 Definition of sheaf and presheaf

Definition 2.2.1. Let X be a space. A *presheaf on X valued in a category $\mathcal{C}$* is a functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$. We often speak of sheaves valued on X , leaving $\mathcal{C}$ implicit, or speak of sheaves valued in $\mathcal{C}$, leaving the space implicit. We denote the category $\text{Fun}(\text{Open}(X)^{\text{op}}, \mathcal{C})$ of presheaves on X valued in $\mathcal{C}$ by $\text{Pshf}_X(\mathcal{C})$. We will often omit $\mathcal{C}$ or X when understood via context. An element $s \in \mathcal{F}(U)$ is called a *section of $\mathcal{F}$ over U* . Sections over X are referred to as *global sections*.

Definition 2.2.2. Let $\mathcal{F}$ be a presheaf on X , and let $p \in U \subseteq X$. The *stalk of $\mathcal{F}$ at p* , denoted by $\mathcal{F}_p$, is the colimit $\text{colim}_{p \in U} \mathcal{F}(U)$, and is explicitly described as the set

$$\{(s, U) \mid p \in U, s \in \mathcal{F}(U)\}$$

modulo $(s, U) \sim (t, V)$ iff there is $W \subseteq U, V$ with $s|_W = t|_W$.

Definition 2.2.3. Let X be a space and $\mathcal{F}$ a presheaf on X valued in a concrete category $\mathcal{C}$. Then $\mathcal{F}$ is a *sheaf* if it satisfies the following axioms:

1. (Locality/identity) If $U \subseteq X$ is open, $\{U_i \subseteq U\}_{i \in I}$ is an open cover, and $s, t \in \mathcal{F}(U)$ are sections such that $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
2. (Gluing) If $U \subseteq X$ is open, $\{U_i \subseteq U\}_{i \in I}$ is an open cover, and $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ is a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

Remark 2.2.4. The locality axiom is equivalent to requiring that sections glue uniquely. To prove this, suppose $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ glue to two sections $s, t \in \mathcal{F}(U)$. Then $s|_{U_i} = s_i = t|_{U_i}$ for all i , and by locality, $s = t$. Conversely, suppose that sections glue uniquely, and let $s, t \in \mathcal{F}(U)$ such that $s|_{U_i} = t|_{U_i}$ for all i . Then define $s_i := s|_{U_i} \in \mathcal{F}(U_i)$, and note that s, t are both given by gluing the s_i 's. By uniqueness, $s = t$. This proves locality.

If $\mathcal{F}$ is a sheaf, then $\mathcal{F}(\emptyset)$ is terminal in $\mathcal{C}$. This allows us to immediately prove that $\mathcal{F}(U \cap V) \cong \mathcal{F}(U) \times \mathcal{F}(V)$ whenever U, V are disjoint. We can generalize sheaves to taking values in arbitrary abelian categories.

Definition 2.2.5. Let $\mathcal{A}$ be an abelian category. A sheaf $\mathcal{F}$ on X valued in $\mathcal{A}$ is a presheaf $\mathcal{F}$ such that for every open $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U , the sequence

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i) \times \mathcal{F}(U_j)$$

is exact. Note that this sequence is an equalizer.

Let X be a space and $U \subseteq X$ an open set. Note that $\text{Open}(U) \subseteq \text{Open}(X)$. Define the restriction sheaf $\mathcal{F}|_U := \mathcal{F}|_{\text{Open}(U)^{\text{op}}}$. We will explore more general restrictions of sheaves with non-open subsets later, but essentially it requires a left adjoint to pushforward along the inclusion map.

Example 2.2.6. Let X a space, $c \in \mathcal{C}$ some object of an abelian or concrete category, and $p \in X$. The *skyscraper sheaf* $i_{p,*}S$ is defined by

$$i_{p,*}S(U) = \begin{cases} S, & p \in U \\ *, & p \notin U \end{cases}.$$

The restriction maps are the identity, except in the case where $p \notin V \subseteq U \ni p$, where the restriction is the unique map $! : S \rightarrow *$.

Example 2.2.7. Let X be a space, $\mathcal{C}$ an abelian or concrete category, and $c \in \mathcal{C}$. The *constant presheaf at c* , denoted $\underline{c}_{\text{pre}}$, is the constant functor $\Delta(c) : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$. In general, this is not a sheaf. In the case where $\mathcal{C}$ is concrete, $\mathcal{V}$ -enriched for some monoidal category $\mathcal{V}$ (usually a cosmos), and each $U \subseteq X$ is an object in some concrete category $\mathcal{S}$ such that the set of locally constant maps in $\mathcal{S}(U, c)$ is in $\mathcal{V}$, then we can define a new sheaf $\mathcal{F}$ to be this set of maps. Such a sheaf is called a constant sheaf, and denoted $\underline{c}$. Generally c is a set and we just consider all locally constant maps $U \rightarrow c$. We will generalize the notion of constant sheaves using sheafification.

Exercise 2.2.8. Let X, Y be spaces. Show that $\text{Top}(-, Y)$ restricted along $\text{Open}(X)^{\text{op}}$ forms a sheaf of sets on X .

Proof. This is immediately functorial since subsets $V \subseteq U$ are precisely inclusion morphisms $V \hookrightarrow U$. We now prove that this is a sheaf. Let $f, g : U \rightarrow Y$ and $\{U_i \subseteq U\}_{i \in I}$ an open cover such that $f|_{U_i} = g|_{U_i}$ for all i . Choose $p \in U$. Then since $\{U_i\}_{i \in I}$ is an open cover, there is $i \in I$ such that $p \in U_i$. Since $f|_{U_i} = g|_{U_i}$, we have $f(p) = g(p)$. Since p is arbitrary, $f = g$. This proves locality. Now let $\{f_i : U_i \rightarrow Y\}_{i \in I}$ be sections satisfying the gluing hypotheses, and $\{U_i \subseteq U\}_{i \in I}$ an open cover. We are forced by remark 2.2.4 to glue these to a section f defined by $f|_{U_i} = f_i$. To show this is well-defined, we must show this definition is independent of the choice of i . Let $p \in U$. Since $\{U_i\}_{i \in I}$ is an open cover, $p \in U_i$ for some i . Let $J \subseteq I$ be the set of all j such that $p \in U_j$. Let $i, j \in J$. It suffices to show $f_i(p) = f_j(p)$, and this is immediate by the gluing hypothesis. ■

Exercise 2.2.9. The previous exercise is a corollary of this. Let $\pi : Y \rightarrow X$ be a map. Show that the sheaf of sections $\Gamma(-, Y)$ is a sheaf.

Proof. If π is not a surjection, then $\Gamma(-, Y) = \Gamma(-)$ is constant at the empty set. This is vacuously a sheaf. Now suppose π is a surjection, and by the axiom of choice it has at least one section.

Most of the proof follows by the above exercise. All that remains to be checked is that restrictions of sections are sections, and gluing sections yields a section. Both are similarly immediate. For the former, let s be a section of π over U , and let $V \subseteq U$ be open. Let $p \in V$. Then $(s|_V)(p) = s(p)$, and thus $(\pi \circ s|_V)(p) = (\pi \circ s)(p) = p$. Thus $s|_V$ is a section over V . Now let $\{s_i \in U_i\}_{i \in I}$ satisfy the gluing hypothesis, where $\{U_i\}_{i \in I}$ is an open cover of some open $U \subseteq X$. We define s by $s|_{U_i} = s_i$. By the logic in the above exercise, this is well-defined as a function. Now we show it is a section. Let $p \in U$; then there is $i \in I$ with $p \in U_i$. We have $(\pi \circ s)(p) = (\pi \circ s_i)(p) = p$ since s_i is a section. Thus s is a section. We did not appeal to continuity here at all; thus this should work in any category whose objects are spaces, and more generally probably works in any concrete category. That is, if Y is a set and we use sections of all functions, or if X, Y have additional structure (eg. topological, smooth, complex manifolds; vector bundles), then this logic should all work. ■

Definition 2.2.10. Let $\pi : X \rightarrow Y$ be a continuous map and $\mathcal{F}$ a presheaf on X . There is a presheaf $\pi_*\mathcal{F}$ on Y , called the *pushforward of $\mathcal{F}$ along π* , defined by $\pi_*\mathcal{F}(U) := \mathcal{F}(\pi^{-1}(U))$. If $\mathcal{F}$ is a sheaf, so is $\pi_*\mathcal{F}$. We will see later that this is a functor $\mathbf{Shf}_X \rightarrow \mathbf{Shf}_Y$.

Exercise 2.2.11. Let $\pi : X \rightarrow Y$ be continuous and $\mathcal{F}$ a sheaf on X . Let $q := \pi(p)$, $p \in X$. There is a morphism $(\pi_*\mathcal{F})_q \rightarrow \mathcal{F}_p$. What is this morphism?

Proof. Recall that $(\pi_*\mathcal{F})_q := \text{colim}_{p \in \pi^{-1}(q)} (\pi_*\mathcal{F})(U)$. This comes with the datum of a cone $c_U^q : (\pi_*\mathcal{F})(U) \rightarrow (\pi_*\mathcal{F})_q$. Note that $(\pi_*\mathcal{F})(U) = \mathcal{F}(\pi^{-1}(U))$, and $p \in \pi^{-1}(U)$. Thus the corresponding cone $c_U^p : \mathcal{F}(U) \rightarrow \mathcal{F}_p$ for $\mathcal{F}_p$ yields maps for all $q \in \pi(U)$, meaning universality induces a unique morphism.

On the level of elements, this is described as follows. A germ (s, U) of $\pi_*\mathcal{F}$ is some $s \in \mathcal{F}(\pi^{-1}(U))$ modulo the relevant relation, where $q \in U$. Thus $p \in \pi^{-1}(U)$, so $(s, \pi^{-1}(U))$ represents a germ in $\mathcal{F}_p$. Thus we obtain a map $[(s, U)] \mapsto [(s, \pi^{-1}(U))]$ which is well-defined. ■

Definition 2.2.12. Let X be a space and $\mathcal{O}_X$ a sheaf of rings on X . A tuple $(X, \mathcal{O}_X)$ is called a *ringed space*, and the sheaf $\mathcal{O}_X$ is called the *structure sheaf*¹.

Definition 2.2.13. Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is a sheaf of abelian groups on X such that for all $U \subseteq X$, the group $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module, which respects restriction in the sense that if $U \subseteq V$, then the diagram

$$\begin{array}{ccc} \mathcal{O}_X(V) \times \mathcal{F}(V) & \longrightarrow & \mathcal{F}(V) \\ \downarrow |_{U \times |U} & & \downarrow |_U \\ \mathcal{O}_X(U) \times \mathcal{F}(U) & \longrightarrow & \mathcal{F}(U) \end{array}$$

commutes, where the horizontal arrows are the scalar action.

Remark 2.2.14. A sheaf of abelian groups is simply a $\mathbb{Z}$ -module. Hence any statements about $\mathcal{O}_X$ -modules apply to sheaves of abelian groups.

The motivating example of an $\mathcal{O}_X$ -module is a vector bundle. Let X be a smooth manifold and $\pi : E \rightarrow X$ a smooth vector bundle. The structure sheaf on X is the sheaf C_X^∞ of smooth maps to $\mathbb{R}$. Given a section $s \in \Gamma(U, E)$ and $f \in C_X^\infty(U)$, there is a section $fs \in \Gamma(U, E)$ defined by $p \mapsto f(p)s_p$. This works since s_p is in an $\mathbb{R}$ -vector space and $f(p) \in \mathbb{R}$, so scalar multiplication yields another vector. Furthermore multiplication by a smooth function is always smooth since multiplication itself is smooth.

¹Generally this is something like C_X^∞ if X is a smooth manifold, or holomorphic functions if it is a complex manifold. We usually choose structure sheaves with meaningful information about the structure of X .

2.3 Morphisms of presheaves and sheaves

Definition 2.3.1. Let $\mathcal{F}, \mathcal{G}$ be presheaves or sheaves. A morphism of presheaves or sheaves is a natural transformation.

Exercise 2.3.2. Let $f : \mathcal{F} \Rightarrow \mathcal{G}$ be a morphism of (pre)sheaves on X , and $p \in X$. Show there is an induced morphism on stalks $\phi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$.

Proof. By functoriality of colimit, this is immediate. To describe the morphism explicitly for concrete categories, let (U, s) be a germ of $\mathcal{F}$ at p . Then $f(s) \in \mathcal{G}(U)$, so $(U, f(s))$ is a germ. This is well-defined since natural transformations commute with restriction morphisms.

More categorically, let $K_p : \mathcal{J} \rightarrow \text{Open}(X)$ be the diagram picking out the full subcategory of $\text{Open}(X)$ spanned by those open sets containing p . Then $\mathcal{F}_p \cong \text{colim}(\mathcal{F} \circ K_p)$. If $f : \mathcal{F} \Rightarrow \mathcal{G}$ is a morphism, then horizontal composition yields a morphism $f \cdot 1 : \mathcal{F} \circ K_p \Rightarrow \mathcal{G} \circ K_p$. Functoriality of $\text{colim} : \mathcal{A}^{\text{Open}(X)} \rightarrow \mathcal{A}$ yields the required morphism. Hence there are *stalkification functors* $(-)_p : \text{Shf}_X(\mathcal{A}) \rightarrow \mathcal{A}$ for any p . ■

Exercise 2.3.3. Show pushforward is a functor.

Proof. Let $\mathcal{F}, \mathcal{G}$ sheaves on X . Let $\pi : X \rightarrow Y$. We define the action on morphisms. Let $f : \mathcal{F} \Rightarrow \mathcal{G}$. We need a map $(\pi_* f)_U : \mathcal{F}(\pi^{-1}(U)) \rightarrow \mathcal{G}(\pi^{-1}(U))$. Note that $\pi^{-1}(U)$ is open, and thus there exists $f_{\pi^{-1}(U)}$. Define

$$(\pi_* f)_U := f_{\pi^{-1}(U)}.$$

This is evidently functorial. ■

Definition 2.3.4. Let $(\mathcal{V}, \otimes)$ be closed symmetric monoidal. Define a tensor product on $\mathcal{P}\text{shf}_X(\mathcal{V})$ pointwise. Define the tensor product $\otimes_{sh}$ of sheaves by sheafifying the presheaf tensor product. Define the internal hom as

$$\text{Hom}_{\mathcal{V}}(\mathcal{F}, \mathcal{G})(U) := \text{Shf}_X(\mathcal{V})(\mathcal{F}|U, \mathcal{G}|U).$$

This construction is an internal hom in all useful situations for this book. Moreover, it is a sheaf if $\mathcal{F}, \mathcal{G}$ are sheaves.

Warning 2.3.5. Sheaf hom does not commute with taking stalks.

2.4 Properties determined at the level of stalks; sheafification

Definition 2.4.1. Let $\mathcal{F}$ be a presheaf on X , and let $U \subseteq X$ be open. A collection of germs $\{f_p\}_{p \in U}$, where $f_p \in \mathcal{F}_p$, is called *compatible* if they are locally germs of a section. Explicitly, for all $q \in U$ there is an open set $q \in V \subseteq U$ and a section $s \in \mathcal{F}(V)$ such that for all $p \in V$, the germ s_p of s at p is

$$s_p = f_p.$$

Proposition 2.4.2. Let $\mathcal{F}$ be a sheaf. Then a collection of compatible germs $\mathfrak{f} := \{f_p\}_{p \in U}$ corresponds to a unique section $s \in \mathcal{F}(U)$ whose germ at p is f_p . We will sometimes call this a collection of compatible germs over U .

Proof. Write s_p for the germ of s at p . Since $\mathfrak{f}$ is compatible by assumption, for each $p \in U$ there is $p \in V_p \subseteq U$ and a section $t \in \mathcal{F}(V_p)$ with $t_q = f_q$ for all $q \in V_p$. Let $a, b \in U$. Let $S \subseteq U$ be a set of points such that the corresponding set of open sets $\{V_p\}_{p \in S}$ is an open cover of U . Let t^p be the section corresponding to V_p . We claim that we can glue $\mathfrak{S} := \{t^p\}_{p \in S}$ together to obtain a section s on U . By the gluing axiom, it suffices to show we can glue together any pair of sections. Let $t^a, t^b \in \mathfrak{S}$. If the corresponding domains V_a, V_b are disjoint then we are done, so assume $V_a \cap V_b$ is nonempty. Choose $p \in V_a \cap V_b$. Then since $t_p^a = f_p = t_p^b$ by assumption, by definition of equality of germs there is an open set $p \in W_p \subseteq V_a \cap V_b$ such that $t^a|_{W_p} = t^b|_{W_p}$. Therefore, t^a agrees with t^b on W_p . We can construct such a W_p for any $p \in V_a \cap V_b$, providing an open cover $\{W_p\}_{p \in V_a \cap V_b}$ of the set, for which the sections t^a and t^b agree on any restriction. By locality and functoriality, $t^a|_{V_a \cap V_b} = t^b|_{V_a \cap V_b}$. Thus $\mathfrak{S}$ glues to a section s with $s|_{V_p} = t_p$, and thus by assumption the germ s_p of s at p is f_p . ■

Remark 2.4.3. Thus in the context of sheaves, compatibility of germs promotes to a global condition: a collection $\mathfrak{f}$ of germs over U is compatible if they are germs of a section s . Note that we have a bijection: given a section over U , we obtain a collection of compatible germs over U . We know already that this function is surjective, so we show that it is injective (the construction in reverse might be thought to be an inverse, but we do not yet know that s is unique). It suffices to show that if two sections have the same germs, they are the same section.

Let $s, t \in \mathcal{F}(U)$ with $s_p = t_p$ for all $p \in U$; here we denote by $(-)_p$ the germ of $(-)$ at p . By locality, it suffices to find an open cover $\{U_i\}_{i \in I}$ of U such that $s|_{U_i} = t|_{U_i}$. Indeed, for each $p \in U$, since $s_p = t_p$, there is an open set $p \in U_p \subseteq U$ with $s|_{U_p} = t|_{U_p}$, by definition of equality of germs. Clearly $\{U_p\}_{p \in U}$ is an open cover of U , so the statement follows.

Thus, sheaves have a bijective correspondence between compatible germs over U and sections over U . This will be useful when constructing sheaves from presheaves on a base.

Exercise 2.4.4. If $f, g : \mathcal{F} \Rightarrow \mathcal{G}$ are morphisms from a presheaf $\mathcal{F}$ to a sheaf $\mathcal{G}$ which induce the same maps on stalks, then $f = g$.

Proof. Write f_p, g_p for the induced maps $\mathcal{F}_p \rightarrow \mathcal{G}_p$. Let $s \in \mathcal{F}(U)$; it suffices to show $f(s) = g(s)$. Since $f_p = g_p$, we have $\prod_{p \in U} f_p = \prod_{p \in U} g_p$. Thus $\{f_p(s_p)\}_{p \in U} = \{g_p(s_p)\}_{p \in U}$. Note that $f(s)_p = f_p(s_p)$ by definition of f_p . Thus $f(s)_p = g(s)_p$ for all $p \in U$. From here we use the same logic as in proposition 2.4.2: equality of germs at all points allows us to cover U by open sets on which $f(s) = g(s)$, allowing us to appeal to locality. ■

Exercise 2.4.5. Show that a morphism of sheaves is an isomorphism if and only if it induces an isomorphism of stalks. Note that this does not say that the stalks are isomorphic, but rather that they are induced by a natural transformation; there may be no way to assemble a collection of isomorphisms of stalks into a natural transformation.

Proof. ($\implies$) Colimit is a functor.

($\impliedby$) Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on X such that $f_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an isomorphism for all $p \in X$. Then there are inverses $g_p : \mathcal{G}_p \rightarrow \mathcal{F}_p$ for each f_p . We must assemble the collection $\{g_p\}_{p \in X}$ into a natural transformation $g : \mathcal{G} \Rightarrow \mathcal{F}$ which provides an inverse to f . Let $U \subseteq X$ be open, and let $s \in \mathcal{G}(U)$. We will show that the collection $\mathfrak{S} := \{g_p(s_p)\}_{p \in U}$ of germs is compatible.

Let $p \in U$, and let $g_p(s_p) \in \mathfrak{S}$. By definition of a germ, there is at least one $t \in \mathcal{F}(V)$ for an appropriate $p \in V \subseteq X$ such that $g_p(s_p) = t_p$. We may assume without loss of generality that $V \subseteq U$, otherwise we take the intersection $V \cap U$. Since $f_p \circ g_p = 1$, it follows that $f_p(t_p) = s_p$.

However, note that $f_p(t_p) = f(t)_p$ by definition. Thus we have $f(t) = s|_V$ by remark 2.4.3. Hence, $s_q = f(t)_q$ for all $q \in V$. It follows that

$$g_q(s_q) = g_p(f(t)_q) = (g_p \circ f_q)(t_q) = t_q.$$

Therefore, for all $q \in V$, we have $g_q(s_q) = t_q$. Thus for any $p \in U$, there exists $V \subseteq U$ and $t \in \mathcal{F}(V)$ such that $g_q(s_q) = t_q$ for all $q \in V$. This is precisely the definition of a collection of compatible germs. Since $\mathcal{F}$ is a sheaf, remark 2.4.3 provides a section $t \in \mathcal{F}(U)$ such that $t_p = g_p(s_p)$ for all p . Define $g_U(s) := t$. This definition is forced in order for g to induce the necessary morphisms on stalks: we require that $g_p(s_p) = g(s)_p$, and indeed this is precisely how it was defined.

The statement now follows immediately from remark 2.4.3. A corollary of that remark is that a section over U is uniquely determined by its germs. Hence, $(f \circ g)_U(s) = s$ if and only if $(f_p \circ g_p)(s_p) = s_p$ for all $p \in U$, which follows by assumption. The same logic proves $g \circ f = 1$. Therefore, $g = f^{-1}$, yielding the desired inverse for f . ■

Definition 2.4.6. Let $U : \mathbf{Shf} \rightarrow \mathbf{Pshf}$ be the forgetful functor. Then sheafification, denoted $(-)^+$, is defined as $(-)^+ \dashv U$.

Proposition 2.4.7. *Sheafification exists.*

Proof. By Freyd's adjoint functor theorem and (co)completeness of $\mathbf{Shf}$, it suffices to show that the forgetful functor preserves small limits. It suffices to show the usual products and equalizers of functors are products and equalizers of sheaves. This is trivial and I avoid working it out. ■

Construction 2.4.8. Let $\mathcal{F}$ be a presheaf. Then $\mathcal{F}^+(U)$ is the set of all collections compatible germs of $\mathcal{F}$ over U . Alternatively, define a space E whose points are germs s_x , where s is any section of $\mathcal{F}$ over any open set in X , and $x \in X$ is any point in that open set. Topologize E by taking open sets to be all sets of the form

$$\{s_x \mid s \in \mathcal{F}(U), x \in U\}$$

for U a fixed open set in X . Then $\mathcal{F}^+ \cong \Gamma(-, E)$.

Exercise 2.4.9. Let $f : \mathcal{F} \rightarrow \mathcal{H}$ be a morphism of sheaves. The following are equivalent.

1. f is a monomorphism.
2. f is injective on stalks.
3. f is injective on open sets.

Important note: the sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X^* \longrightarrow 0$$

is exact.

2.5 Recovering sheaves from a “sheaf on a base”

Definition 2.5.1. Let X be a space with a base $\mathcal{B}$. A *presheaf* $\mathcal{F}$ on a base of X with values in $\mathcal{C}$ is a functor $\mathcal{B}^{\text{op}} \rightarrow \mathcal{C}$, where $\mathcal{B}$ is viewed as a poset category. A *sheaf* on a base is a presheaf on a base which satisfies the analogues of locality and gluing for only elements of the base.

Theorem 2.5.2. *Let $\mathcal{F}$ be a sheaf on a base $\mathcal{B} = \{B_i\}_{i \in I}$ of a space X . Then there is a sheaf $\mathcal{F}$ extending $\mathcal{F}$ which is unique up to unique isomorphism.*

Proof. The obvious answer is to use compatible germs. Define the stalk of a sheaf on a base as $\mathcal{F}_p = \text{colim}_{p \in B} \mathcal{F}(B)$. Define $\mathcal{F}(U)$ to be all collections of compatible germs over U , which in this case must be defined using the base. There is an obvious natural isomorphism from $\mathcal{F}$ to the subfunctor $\mathcal{F}|_{\mathcal{B}}$ of $\mathcal{F}$. ■

Exercise 2.5.3. Let $\{U_i\}_{i \in I}$ be a cover of some space X , and let $\mathcal{F}_i$ be sheaves on U_i which are compatible in the sense that there are isomorphisms

$$\varphi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \cong \mathcal{F}_j|_{U_i \cap U_j}$$

which satisfy the cocycle condition

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik},$$

and such that $\varphi_{ii} = 1$. Then there is a sheaf $\mathcal{F}$ which is unique up to isomorphism and satisfies $\mathcal{F}|_{U_i} = \mathcal{F}_i$.

2.6 Sheaves of abelian groups, and $\mathcal{O}_X$ -modules, form abelian categories

I believe I already commented on how all one must prove for $\text{Shf}(\mathcal{A})$ to be abelian when $\mathcal{A}$ is abelian is stability under colimits. Sheafification provides the relevant colimits. Hence, the only new thing here is $\mathcal{O}_X$ -modules, but the proof for those simply requires showing that $\mathcal{O}_X$ -modules are stable under the sheaf (co)limits. This is immediate since it reduces to the case of (co)limits of modules, which are well-understood. This section then goes on to note that a lot of functors are left- or right-exact, but they are all right- or left-adjoint, respectively. Thus, I will skip this section.

2.7 The inverse image sheaf

Definition 2.7.1. Let $\pi : X \rightarrow Y$ be continuous. Then define the functor $\pi^{-1} : \text{Shf}_Y \rightarrow \text{Shf}_X$ as $\pi^{-1} \dashv \pi_*$.

It is simple to prove that π_* preserves limits. Thus, π^{-1} exists by Freyd's adjoint functor theorem (see [ML98]). The explicit construction is as follows.

Construction 2.7.2. Let $\mathcal{G}$ be a presheaf. Define $(\pi_{\text{pre}}^{-1}\mathcal{G})(U) := \text{colim}_{\pi(U) \subseteq V} \mathcal{G}(V)$. If $\mathcal{G}$ is a sheaf, define $\pi^{-1}\mathcal{G} := (\pi_{\text{pre}}^{-1}\mathcal{G})^+$.

The inverse image sheaf provides us a way to restrict sheaves to non-open subsets. Given X a space, $\mathcal{F}$ a sheaf on X , and $A \subseteq X$ with inclusion morphism $i : A \subseteq X$, define $\mathcal{F}|_A$ to be $i^{-1}\mathcal{F}$. It is simple to show that in the case where A is open, these notions agree. Fix an open set $\mathcal{U} \subseteq X$, and consider the restriction $\mathcal{F}|_{\mathcal{U}}$. We have that

$$(i^{-1}\mathcal{F})(U) := \text{colim}_{U \subseteq V \text{ open in } \mathcal{U}} \mathcal{F}(V).$$

Here we have used $i^{-1}(U) = U$ to simplify the colimit. Note that this colimit is taken over a poset category. Since the functor is contravariant, we remark that the limit of this diagram (hence the colimit in the opposite category) is a lower bound for the $\subseteq$ relation. It's easy to show that $\mathcal{F}$ preserves this colimit (not all colimits though), and thus we have that $(i^{-1}\mathcal{F})(U) \cong \mathcal{F}(U)$ by left-adjointness of sheafification.

Remark 2.7.3. The inverse image functor is an exact functor for sheaves taking values in abelian categories. The same logic yields an exact functor π^{-1} from $\mathcal{O}_Y$ -modules on Y to $(\pi^{-1}\mathcal{O}_Y)$ -modules on X . This is easy to show: left-adjointness yields right exactness, and left-exactness follows since colimits over filtered index sets are exact.

Definition 2.7.4. Let

$$\begin{array}{ccc} W & \xrightarrow{\beta'} & X \\ \alpha' \downarrow & & \downarrow \alpha \\ Y & \xrightarrow{\beta} & Z \end{array}$$

commute, and let $\mathcal{F}$ be a sheaf on X . Define the push-pull map of sheaves on Y as follows. Transpose the identity on $(\beta')^{-1}\mathcal{F}$ under $(\beta')^{-1} \dashv \beta'_*$ and apply α_* to obtain $\alpha_*\mathcal{F} \rightarrow \alpha_*(\beta'_*)(\beta')^{-1}\mathcal{F}$. Apply $\alpha\beta' = \beta\alpha'$ and transpose back under the adjunction to get the desired map.

Definition 2.7.5. Let $\mathcal{F}$ be a sheaf. Define the support $\text{Supp}(s)$ of a global section s of $\mathcal{F}$ (one can define this more locally if one desires) as the set of points where s has nonzero germ:

$$\text{Supp}(s) := \{p \in X \mid s_p \neq 0 \in \mathcal{F}_p\}.$$

If $\mathcal{F}$ is valued in an abelian category, we define the support of the sheaf $\mathcal{F}$ as

$$\text{Supp } \mathcal{F} := \{p \in X \mid \mathcal{F}_p \neq 0\}.$$

Part II

Schemes

Chapter 3

Toward affine schemes: the underlying set, and topological space

3.1 Toward schemes

The idea behind a scheme is as follows. Given a geometric space, such as a smooth, complex, or real analytic manifold, we can usually study this space using particular functions on that space (for the given examples, $C_M^\infty, C_M^\omega, \mathcal{O}_X$, respectively). These are usually taken to be the structure sheaf of the space. The idea of a scheme is to generalize such a notion of a geometric space in such a way that the tools of algebra can be more readily applied. Thus we will define a scheme to be a particular class of ringed spaces.

Example 3.1.1. Let M be a smooth manifold, and recall that (M, C_M^∞) is a ringed space. By continuity of smooth maps, the evaluation of a germ f_p at p is well-defined, so evaluation yields a map

$$\varepsilon_p : C_{M,p}^\infty \rightarrow \mathbb{R}.$$

This map is clearly surjective. To show this, choose a chart U containing p , choose a closed neighborhood $p \in A \subseteq U$ which exists by $U \cong V \subseteq \mathbb{R}^n$, and choose a bump function f for A supported in U . Then $\varepsilon_p(kf) = k$ for any $k \in \mathbb{R}$. The first isomorphism theorem yields $C_{M,p}^\infty / \text{Ker } \varepsilon_p \cong \mathbb{R}$. Since $\mathbb{R}$ is a field and fields have no nontrivial ideals, $\text{Ker } \varepsilon_p := \mathfrak{m}_{M,p}$ is a maximal ideal of the stalk $C_{M,p}^\infty$.

We could choose to take the reverse approach, defining a smooth manifold to be a space with a sheaf of rings $\mathcal{O}_X$, and a cover by open sets that are isomorphic to balls in $\mathbb{R}^n$ (wlog we can take these balls centered at the origin). The manifold is simply given by gluing the cover together. Algebraically, elements of the cover translate to the notion of an *affine scheme*. We can also describe morphisms of manifolds in this language, by requiring that smooth functions pull back to smooth functions. This is sufficient: given $f : M \rightarrow N$ continuous such that all smooth $g : U \rightarrow \mathbb{R}$ pull back to a smooth map $g \circ f|_{f^{-1}(U)} : f^{-1}(U) \rightarrow \mathbb{R}$, it follows that f is smooth by letting g be a component of local coordinates. The book writes $f^\#$ for this morphism $C_N^\infty \rightarrow \pi_* C_M^\infty$, which I hate so unless there is a conflict of notation, I will write f^* . Note that the pushforward here is so that this becomes a sheaf morphism; it essentially just makes evaluation of C_M^∞ default to an open set of the form $f^{-1}(U)$.

Much of this language also applies if one is interested in the more general notion of a topological space; the correct structure sheaf here is clearly $C^0(-, \mathbb{R})$. The point here is that geometric spaces can be viewed as topological spaces with a particular structure sheaf (usually a subsheaf of C^0). Affine schemes also fit into this family.

3.2 The underlying set of an affine scheme

Definition 3.2.1. Let A be a ring. Define $\text{Spec } A$ to be the set of prime ideals of A . Given a prime ideal $\mathfrak{p} \subseteq A$, we will denote $\mathfrak{p}$ by $[\mathfrak{p}]$ when considered as an element of A . An element $a \in A$ is called a *function on $\text{Spec } A$* , and their *value* at the point $\mathfrak{p}$ will be $a \pmod{\mathfrak{p}}$. Note that this is only well-defined if $a([\mathfrak{p}]) \in A/\mathfrak{p}$, meaning for a to be formally viewed as a function, we must define

$$a : \text{Spec } A \rightarrow \bigoplus_{\mathfrak{p} \text{ prime}} \frac{A}{\mathfrak{p}}.$$

Note that if $a \in \mathfrak{p}$, then the value of a at $[\mathfrak{p}]$ is 0. We can add or multiply functions on $\text{Spec } A$, which is well-defined since the projections $A \rightarrow A/\mathfrak{p}$ are ring homomorphisms. If A is generated over another ring by elements $X = \{x_1, \dots, x_r\}$, then the set X is referred to as *coordinates*.

Example 3.2.2. The complex affine line is defined as $\mathbb{A}_{\mathbb{C}}^1 := \text{Spec } \mathbb{C}[x]$. Since $\mathbb{C}[x]$ is an integral domain, 0 is prime. Additionally, $(x - a)$ is maximal for all $a \in \mathbb{C}$, since quotienting out by it just yields $\mathbb{C}$, a field. Hence all ideals $(x - a)$ are prime. One can show that there are no other prime ideals by dividing polynomials and getting some contradictions.

Thus we can geometrically interpret affine space. For all $a \in \mathbb{C}$, we get a point $[(x - a)]$ in $\text{Spec } \mathbb{C}[x]$, but that leaves an extra point $[(0)]$ which is not accounted for by these points. However, $[(0)]$ is included in any of the other ideals, so we could imagine it living somewhere within any of them. More formally, we associate it with the line passing through all other points, and call it the generic point of this line.

Interestingly, in example 3.2.2, if I have a polynomial f interpreted as a function on $\mathbb{A}_{\mathbb{C}}^1$, then its value at $[(x - a)]$ is simply $f(a)$. The proof is very messy algebra. We can also talk about rational functions $(x - 3)^3/(x - 2)$, which are not functions in the sense of elements of $\mathbb{C}[x]$, but can be defined more generally as an algebraic function that exists everywhere except $x = 2$. We will later define it as a section of $\mathcal{O}_{\mathbb{A}_{\mathbb{C}}^1}(\mathbb{A}_{\mathbb{C}}^1 \setminus \{2\})$, and we will also later be able to say that this function has a triple zero at $x = 3$ in this language.

Example 3.2.3. Let k be an algebraically closed field. Then $\mathbb{A}_k^1 := \text{Spec } k[x]$. All of the same discussion as before applies.

Example 3.2.4. Consider $\text{Spec } \mathbb{Z}$. The prime ideals are 0 and (p) for p prime. The same picture as before works, but with a more discrete space. We can also discuss functions on this space. The function 100 has value $100[(3)] = 1$, and a double zero $100[(2)] = 0$. There is a rational function $27/4$ defined away from (2) . It has a “double pole” at (2) , and a triple zero at (3) . Its value at (5) is 3. We will make this discussion more precise over time.

Example 3.2.5. The real affine line $\mathbb{A}_{\mathbb{R}}^1$ consists of the ideals generated by 0, linear functions, and irreducible quadratics. Linears and irreducible quadratics generate maximal ideals.

We could continue with examples. An interesting one is that $\text{Spec } \mathbb{F}_p[x]$ has infinitely many points. In fact, $\text{Spec } k[x]$ has infinitely many points for all fields k .

Example 3.2.6. The complex affine plane is $\mathbb{A}_{\mathbb{C}}^2 := \text{Spec } \mathbb{C}[x, y]$. As before, we could replace $\mathbb{C}$ with any algebraically closed field. As before, 0 is prime. For any $(a, b) \in \mathbb{C}^2$, the ideal $(x - a, y - b)$ is prime with $\mathbb{C}[x, y]/(x - a, y - b) \cong \mathbb{C}$ via $f \mapsto f(a, b)$. Any irreducible polynomial also determines a prime ideal. This determines all prime ideals of $\mathbb{C}[x, y]$. To picture $\mathbb{A}_{\mathbb{C}}^2$, first identify $(x - a, y - b)$ with the point (a, b) . We think of irreducibles $f(x, y)$ as determining their graph, so we identify the

ideal generated by $f(x, y) = y - x^2$ with the parabola $y = x^2$. We then finally set $[(0)]$ as living behind everything; everywhere yet nowhere in particular. Note that maximal ideals correspond to the smallest points, and this geometric picture appears to preserve the $\subseteq$ relation.

Definition 3.2.7. Let A be a ring. Then the n -dimensional affine space over A is defined as $\text{Spec } A[x_1, \dots, x_n]$.

Hilbert's nullstellensatz is useful here.

Theorem 3.2.8 (Weak nullstellensatz). *Let k be an algebraically closed field. Then the maximal ideals of $k[x_1, \dots, x_n]$ are precisely those ideals of the form $(x_1 - a_1, \dots, x_n - a_n)$, for $a_i \in k$.*

Theorem 3.2.9 (Nullstellensatz, Zariski). *If k is a field, then every maximal ideal of $k[x_1, \dots, x_n]$ has residue field a finite extension of k . Alternatively, any field extension of k that is finitely generated as a k -algebra is also finitely generated as a k -vector space.*

The prime ideals of the form in theorem 3.2.8 are geometrically interpreted as the (0-dimensional) points $(a_1, \dots, a_n)$. Over an algebraically closed field, we also have an n -dimensional point $[0]$. Given an irreducible polynomial $f(x)$ in the n -variables, we view $[(f)]$ as its graph, which is an $(n - 1)$ -dimensional point. This is the best classification we can give algebraically. There are more prime ideals, corresponding to the other dimensions. For example, (x_1, x_2) is an $(n - 2)$ -dimensional point. Thus continuing this classification in any meaningful or reasonable way is a geometric question, not an algebraic one.

Recall that there is a bijective correspondence between prime ideals of A/I and prime ideals of A containing I . Thus we have inclusions $\text{Spec } A/I \subseteq \text{Spec } A$. Suppose A is a $\mathbb{C}$ -algebra generated by $x_1, \dots, x_n$ modulo relations $f_i(x) = 0$, $i = 1, \dots, r$. We can naturally extract a picture of $\text{Spec } A$ as a subset of $\mathbb{A}_{\mathbb{C}}^n$ using this picture. The zero-dimensional points are given by the solutions to $f_1 = \dots = f_r = 0$. The higher dimensional points are more complex.

Let $f : R \rightarrow S$ be a map of rings. Then $f^{-1}(\mathfrak{p})$ is prime for any prime ideal $\mathfrak{p} \subseteq S$, so there is an induced map $\text{Spec } f : \text{Spec } S \rightarrow \text{Spec } R$. In particular, Spec is a functor $\text{Ring}^{\text{op}} \rightarrow \text{Set}$.

Let $r \in R$ such that $r[\mathfrak{p}] = 0$ for all $\mathfrak{p}$. It does not follow that $r = 0$. For example, consider $k[\varepsilon]/(\varepsilon^2)$. Note that since $\varepsilon^2 = 0$, we have $\varepsilon \in \mathfrak{p}$ for any prime ideal, and thus ε is zero as a function on prime ideals, but is not zero as an element of R . An element r with $r^n = 0$ is called *nilpotent*. To get rid of this issue, one might think that we must get rid of zero-divisors.

Definition 3.2.10. Let R be a ring. If $r \in R$ is such that $r^n = 0$ for some n , then r is called *nilpotent*. The collection of nilpotents of R is denoted $\mathfrak{R}(R) = \mathfrak{R}$, and is called the *nilradical*. It is obviously an ideal.

Theorem 3.2.11. *Let R be a ring. Then $\mathfrak{R}(A)$ is the intersection of all prime ideals of R .*

3.3 generic points

skip

3.4 The underlying topological space of an affine scheme

Definition 3.4.1. Let A be a ring. Given $S \subseteq A$, define the *vanishing set* of S by

$$V(S) := \{[\mathfrak{p}] \in \text{Spec } A \mid S \subseteq \mathfrak{p}\}.$$

The *Zariski topology* on $\text{Spec } A$ is the topology where the closed sets are precisely the sets $V(S)$ for all subsets $S \subseteq A$.

The Zariski topology makes precise the intuition that we have been considering so far. To illustrate this, let $\mathfrak{p} \subseteq \mathfrak{q}$ be an inclusion of prime ideals. Earlier, we graphically identified a prime ideal with its zero locus $V(\mathfrak{p})$. Note that $S_1 \subseteq S_2$ implies $V(S_2) \subseteq V(S_1)$. This means that graphically, the point $\mathfrak{p}$ should contain the point $\mathfrak{q}$. Since points are usually closed sets (in all Hausdorff spaces), this intuition makes sense.

Definition 3.4.2. Let $I \subseteq A$ be an ideal. Define the *radical* $\sqrt{I}$ as

$$\sqrt{I} := \{r \in A \mid r^n \in I \text{ for some } n \in \mathbb{Z}_{>0}\}.$$

The radical of an ideal is an ideal, and has the properties that $\sqrt{\sqrt{I}} = \sqrt{I}$, and it commutes with finite intersections:

$$\sqrt{\bigcap_{i=1}^n I_i} = \bigcap_{i=1}^n \sqrt{I_i}.$$

Additionally, $\sqrt{I}$ is the intersection of all prime ideals containing I .

Let $\phi : A \rightarrow B$ be a morphism of rings. Then the preimage map $\phi^* : \text{Spec } B \rightarrow \text{Spec } A$ is continuous.

3.5 A base of the Zariski topology on $\text{Spec } A$: distinguished open sets

Definition 3.5.1. Let A be a ring and $f \in A$. Define the *distinguished open set*

$$D(f) := \{[\mathfrak{p}] \in \text{Spec } A \mid f \notin \mathfrak{p}\}$$

as the nonvanishing locus of f . Hence, distinguished open sets are the compliments of vanishing loci of singletons.

These form a base for the topology on $\text{Spec } A$.

Remark 3.5.2. He never defined what $\text{Spec } A_f$ meant, but now he notes that $D(f) \cong \text{Spec } A_f$ as spaces, so I will take this as a definition.

An important fact is that

$$\bigcup_{i \in J} D(f_i) = \text{Spec } A \text{ iff } \{f_i\}_{i \in J} = A.$$

Another important fact is that an equivalent statement to $\{f_i\}_{i \in J} = A$ is that there is a partition of unity: there are $\{a_i\}_{i \in J} \subseteq A$, all but finitely many 0, such that $\sum_i a_i f_i = 1$.

3.6 Topological and Noetherian properties

Note that $\text{Spec} : \text{Ring}^{\text{op}} \rightarrow \mathcal{T}\text{op}$. It can be shown that Spec preserves finite coproducts, i.e.

$$\text{Spec} \prod_{1 \leq i \leq n} A_i \cong \prod_{1 \leq i \leq n} \text{Spec } A_i.$$

Note that this implies $\text{Spec } A$ *cannot be connected* if it can be reduced to a product of nonzero rings. This statement turns out to be an if and only if.

Recall that a topological space is called *irreducible* if it is not the union of two proper *closed* subsets. This is a very strange notion, since these subsets *need not be disjoint*. For example, $[0, 1]$ is not irreducible. A useful and equivalent definition is that any two nonempty open subsets of X must intersect. An example of such a space is the cofinite topology on a finite set; the closed sets are too small for the space to be a union of finitely many proper closed sets. Many spectra are irreducible since they have similar topologies to the cofinite topology. For example, $\mathbb{A}_{\mathbb{C}}^2$ is irreducible.

Any nonempty open set in an irreducible space is dense. The spectrum of any integral domain is irreducible. All irreducible spaces are connected.

The book refers to compactness as “quasicompactness”. I hate this.

Definition 3.6.1. Let X be a space. A *closed point* is a point whose singleton is closed.

Given a ring A , the closed points of $\text{Spec } A$ are precisely the maximal ideals. Given a space X and $x, y \in X$, we say y is a *generalization* of x and x is a *specialization* of y if $x \in \overline{\{y\}}$.

Definition 3.6.2. A point $p \in X$ for a space X is a *generic point* for a closed subset K if $\overline{\{p\}} = K$.

Note that the smallest open set containing $\mathfrak{p} \in \text{Spec } A$ is $V(\mathfrak{p})$. This contains $V(\mathfrak{m})$ for any and all maximal ideals $\mathfrak{m}$ such that $\mathfrak{p} \subseteq \mathfrak{m}$. The same can be said for any other prime ideals containing $\mathfrak{p}$. Thus the “generalized points” we were talking about are *generic points*.

A topological space X is *Noetherian* if every chain $Z_1 \supseteq Z_2 \supseteq \cdots$ of closed sets stabilizes. Then every closed set is a finite union of irreducible closed sets, none of which contain each other.

Theorem 3.6.3 (Hilbert). *Let A be Noetherian. Then so is $A[x]$.*

If A is Noetherian, then $\text{Spec } A$ is Noetherian as a space. The converse need not hold.

3.7 The function I , taking subsets of $\text{Spec } A$ to ideals of A

Definition 3.7.1. Given a ring A and a subset $S \subseteq \text{Spec } A$, define the ideal $I(S)$ of A to be

$$I(S) := \bigcap_{\mathfrak{p} \in S} \mathfrak{p}.$$

Note that the empty intersection is 0 since $\cap$ is a limit, so this is always an ideal.

I is an inclusion reversing property such that $I(\overline{S}) = I(S)$, for $\overline{S}$ the closure. One can also prove the following:

- $V(I(S)) = \overline{S}$;
- $I(V(J)) = \sqrt{J}$ for ideals J .

Theorem 3.7.2. *The functions V, I yield an inclusion-reversing bijection between closed subsets of $\text{Spec } A$ and radical ideals of A .*

Chapter 4

The structure sheaf, and the definition of schemes in general

4.1 The structure sheaf of an affine scheme

The intuition for $\mathcal{O}_{\text{Spec } A}$ will be that $\mathcal{O}_{\text{Spec } A}(U)$ contains all algebraic functions on the open set U . A good motivation is algebraic functions on $\mathbb{C}^n$. For example, on the nonvanishing locus $D(xy)$, we would expect $(3x^4 + y + 4)/x^7y^3$ to be an algebraic function.

Definition 4.1.1. Let A be a ring, and $f \in A$ a function. Define $\mathcal{O}_{\text{Spec } A}(D(f))$ to be the localization $A[S^{-1}]$, for S the multiplicative set of all functions $g \in A$ that do not vanish outside of $V(f)$. Equivalently, $D(f) \subseteq D(g)$.

Theorem 4.1.2. The presheaf $\mathcal{O}_{\text{Spec } A}$ is a sheaf on a base, and thus extends to a sheaf on $\text{Spec } A$.

Note that the presheaf structure is defined as follows. Let $f, g \in A$ with $D(f) \subseteq D(g)$. Thus, the nonvanishing locus of f is smaller than that of g , meaning $f \notin [\mathfrak{p}]$ implies $g \notin [\mathfrak{p}]$ for a prime ideal $\mathfrak{p}$. We want to define a restriction $\mathcal{O}_{\text{Spec } A}(D(g)) \rightarrow \mathcal{O}_{\text{Spec } A}(D(f))$. Write S_f, S_g for the multiplicative sets of functions with larger nonvanishing loci than f, g , respectively. If $k \in S_g$, then $D(g) \subseteq D(k)$. But $D(f) \subseteq D(g)$ implies $D(f) \subseteq D(k)$, and thus $k \in S_f$. Therefore $S_g \subseteq S_f$. The canonical quotient map $\pi_f : A \rightarrow A[S_f^{-1}]$ thus inverts all elements in S_g , and by universality, factors through the quotient $\pi_g : A \rightarrow A[S_g^{-1}]$ as the unique dashed arrow rendering the diagram

$$\begin{array}{ccc} A & \xrightarrow{\pi_f} & A[S_f^{-1}] \\ \pi_g \downarrow & \nearrow & \\ A[S_g^{-1}] & & \end{array}$$

commutative. This is the restriction morphism $\mathcal{O}_{\text{Spec } A}(D(g)) \rightarrow \mathcal{O}_{\text{Spec } A}(D(f))$. We can explicitly describe this map by viewing $A[S_f^{-1}]$ as A adjoined with formal inverses of S . If $k \in S_g$, then $k \in S_f$, implying that $1/k \in A[S_g^{-1}]$ also exists as $1/k \in A[S_f^{-1}]$. Thus this map can be thought of as the inclusion of a subset $A[S_g^{-1}] \subseteq A[S_f^{-1}]$.

Definition 4.1.3. Let M be an A -module. Then the following construction defines an $\mathcal{O}_A$ -module $\widetilde{M}$. Define $\widetilde{M}(D(f))$ to be the localization of M at the multiplicative set of all functions that do not vanish outside of $V(f)$, and define restriction maps in the same way.

4.2 ?

4.3 Definition of schemes

Definition 4.3.1. An isomorphism of ringed spaces $(X, \mathcal{O}_X) \cong (Y, \mathcal{O}_Y)$ is a homeomorphism $\pi : X \cong Y$ and an isomorphism of sheaves $f : \mathcal{O}_Y \cong \pi_* \mathcal{O}_X$.

Definition 4.3.2. An *affine scheme* is a ringed space that is isomorphic to $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ for some A . A *scheme* $(X, \mathcal{O}_X)$ is a ringed space that is locally an affine scheme. The topology on X is called the Zariski topology. The category of schemes is defined to be the full subcategory of ringed spaces spanned by schemes. We call sections $\mathcal{O}_X(U)$ “functions on U ”.

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